

Process Credentials

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Process User & Group IDs

Process User & Group IDs

- real user and group ID

Process User & Group IDs

- real user and group ID
- effective user and group ID

Process User & Group IDs

- real user and group ID
- effective user and group ID
- saved set-user-ID and saved set-group-ID

Process User & Group IDs

- real user and group ID
- effective user and group ID
- saved set-user-ID and saved set-group-ID
- supplementary group IDs

Real User and Group ID

```
$ cat /etc/passwd
root:x:0:0:System administrator:/root:/run/current-system/sw/bin/bash
ali:x:1000:100::/home/ali:/run/current-system/sw/bin/bash
tor:x:35:35:Tor Daemon User:/var/lib/tor:/run/current-system/sw/bin/nologin
gdm:x:132:132:GDM user:/run/gdm:/run/current-system/sw/bin/nologin
nm-openvpn:x:217:217::/var/empty:/run/current-system/sw/bin/nologin
$
```

Supplementary Groups

```
$ cat /etc/group | grep ali
wheel:x:1:ali
docker:x:131:ali
$ id
uid=1000(ali) gid=100(users) groups=100(users),1(wheel),131(docker)
```


Effective User and Group ID

```
int fd;
uid_t ruid, euid, suid;
char buf[BUFSIZ];

getresuid(&ruid, &euid, &suid);
printf("ruid: %d, euid: %d, suid: %d\n", ruid, euid, suid);

fd = open(argv[1], O_RDONLY);
if (fd == -1)
    err(-1, "failed to open %s", argv[1]);

read(fd, buf, sizeof(buf));
puts(buf);

// $ cc ./vidcat.c -o vidcat
// $ ./vidcat /etc/shadow
// ruid: 1000, euid: 1000, suid: 1000
// vidcat: failed to open /etc/shadow: Permission denied
```

Effective User and Group ID

#

Effective User and Group ID

```
# ls -ltrh /tmp/tests/  
total 8.0K  
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt  
-rw-r----- 1 tor tor   30 May 13 22:30 tor-file.txt  
#
```

Effective User and Group ID

```
# ls -ltrh /tmp/tests/
total 8.0K
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt
-rw-r----- 1 tor tor   30 May 13 22:30 tor-file.txt
# sudo -u ali ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
#
```

Effective User and Group ID

```
# ls -ltrh /tmp/tests/
total 8.0K
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt
-rw-r----- 1 tor tor   30 May 13 22:30 tor-file.txt
# sudo -u ali ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
# sudo -u ali ./vidcat /tmp/tests/ali-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
just for ali user and group
#
```

Effective User and Group ID

```
# ls -ltrh /tmp/tests/
total 8.0K
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt
-rw-r----- 1 tor tor   30 May 13 22:30 tor-file.txt
# sudo -u ali ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
# sudo -u ali ./vidcat /tmp/tests/ali-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
just for ali user and group
# sudo -u tor ./vidcat /tmp/tests/tor-file.txt
ruid: 35, euid: 35, suid: 35
rgid: 35, egid: 35, sgid: 35
just for tor user and group
#
```

Effective User and Group ID

```
# ls -ltrh /tmp/tests/
total 8.0K
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt
-rw-r----- 1 tor tor   30 May 13 22:30 tor-file.txt
# sudo -u ali ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
# sudo -u ali ./vidcat /tmp/tests/ali-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
just for ali user and group
# sudo -u tor ./vidcat /tmp/tests/tor-file.txt
ruid: 35, euid: 35, suid: 35
rgid: 35, egid: 35, sgid: 35
just for tor user and group
# sudo -u tor ./vidcat /tmp/tests/ali-file.txt
ruid: 35, euid: 35, suid: 35
rgid: 35, egid: 35, sgid: 35
vidcat: failed to open /tmp/tests/ali-file.txt: Permission denied
```

Effective User and Group ID

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total 8.0K  
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt  
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#
```


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total 8.0K
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt
-rw-r----- 1 tor tor   30 May 13 22:30 tor-file.txt

# sudo -u ali -g tor ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 35, egid: 35, sgid: 35
just for tor user and group

#
```

Effective User and Group ID

```
# ls -ltrh /tmp/tests/
```

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total 8.0K
```

```
-rw-r----- 1 ali users 30 May 13 22:29 ali-file.txt
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```
# sudo -u ali -g tor ./vidcat /tmp/tests/tor-file.txt
```

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ruid: 1000, euid: 1000, suid: 1000
```

```
rgid: 35, egid: 35, sgid: 35
```

```
just for tor user and group
```

```
# sudo -u tor -g users ./vidcat /tmp/tests/tor-file.txt
```

```
ruid: 35, euid: 35, suid: 35
```

```
rgid: 100, egid: 100, sgid: 100
```

```
just for tor user and group
```

```
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```
total 8.0K
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ruid: 1000, euid: 1000, suid: 1000
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rgid: 35, egid: 35, sgid: 35
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just for tor user and group
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```
# sudo -u tor -g users ./vidcat /tmp/tests/tor-file.txt
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ruid: 35, euid: 35, suid: 35
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rgid: 100, egid: 100, sgid: 100
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just for tor user and group
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```

```
ruid: 1000, euid: 1000, suid: 1000
```

```
rgid: 100, egid: 100, sgid: 100
```

```
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
```

set user or group ID on execution (s)
chmod ug+s filename

Set-User-ID and Set-Group-ID

#

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ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
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```

Set-User-ID and Set-Group-ID

```
# sudo -u ali -g users ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
# chown ali:tor vidcat && chmod g+s vidcat
#
```

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# sudo -u ali -g users ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 100, sgid: 100
vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
# chown ali:tor vidcat && chmod g+s vidcat
# sudo -u ali -g users ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 35, sgid: 35
just for tor user and group
#
```


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ruid: 1000, euid: 1000, suid: 1000
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vidcat: failed to open /tmp/tests/tor-file.txt: Permission denied
# chown ali:tor vidcat && chmod g+s vidcat
# sudo -u ali -g users ./vidcat /tmp/tests/tor-file.txt
ruid: 1000, euid: 1000, suid: 1000
rgid: 100, egid: 35, sgid: 35
just for tor user and group
# chown tor:users vidcat && chmod u+s vidcat
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```

Set-User-ID and Set-Group-ID

```
# sudo -u ali -g users ./vidcat /tmp/tests/tor-file.txt
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rgid: 100, egid: 35, sgid: 35
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# chown tor:users vidcat && chmod u+s vidcat
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rgid: 100, egid: 100, sgid: 100
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Set-User-ID and Set-Group-ID

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ruid: 1000, euid: 35, suid: 35
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just for tor user and group
```

Retrieving real and effective IDs

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 - root: changes real, effective, and saved GID (all three)
 - non-root: changes ONLY effective GID (to real OR saved GID)
- `int seteuid(uid_t euid);`
- `int setegid(gid_t egid);`
- `int setreuid(uid_t ruid, uid_t euid);`

Modifying effective IDs

- `int setuid(uid_t uid);`
 - root: changes real, effective, and saved UID (all three)
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- `int setgid(gid_t gid);`
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 - non-root: changes ONLY effective GID (to real OR saved GID)
- `int seteuid(uid_t euid);`
- `int setegid(gid_t egid);`
- `int setreuid(uid_t ruid, uid_t euid);`
- `int setregid(gid_t rgid, gid_t egid);`

Effective User and Group ID

```
pw = getpwnam(argc > 1 ? argv[1] : "root");  
if (!pw || !(sp = getsppnam(pw->pw_name)))  
    errx(-1, "user/shadow failed");
```

Effective User and Group ID

```
pw = getpwnam(argc > 1 ? argv[1] : "root");
if (!pw || !(sp = getspnam(pw->pw_name)))
    errx(-1, "user/shadow failed");

pass = getpass("Password: ");
if (!pass || strcmp(crypt(pass, sp->sp_pwdp), sp->sp_pwdp))
    errx(-1, "authentication failed");
```

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```
pw = getpwnam(argc > 1 ? argv[1] : "root");
if (!pw || !(sp = getspnam(pw->pw_name)))
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pass = getpass("Password: ");
if (!pass || strcmp(crypt(pass, sp->sp_pwdp), sp->sp_pwdp))
    errx(-1, "authentication failed");

setenv("HOME", pw->pw_dir, 1);
setenv("USER", pw->pw_name, 1);
```

Effective User and Group ID

```
pw = getpwnam(argc > 1 ? argv[1] : "root");
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    errx(-1, "user/shadow failed");

pass = getpass("Password: ");
if (!pass || strcmp(crypt(pass, sp->sp_pwdp), sp->sp_pwdp))
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setenv("HOME", pw->pw_dir, 1);
setenv("USER", pw->pw_name, 1);

if (setgid(pw->pw_gid))
    err(-1, "failed to set GID");
if (setuid(pw->pw_uid))
    err(-1, "failed to set UID");
```


Effective User and Group ID

```
pw = getpwnam(argc > 1 ? argv[1] : "root");
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setenv("HOME", pw->pw_dir, 1);
setenv("USER", pw->pw_name, 1);

if (setgid(pw->pw_gid))
    err(-1, "failed to set GID");
if (setuid(pw->pw_uid))
    err(-1, "failed to set UID");

execlp(pw->pw_shell, pw->pw_shell, NULL);
// # cc su.c -o su -lcrypt
// # chown root:root su && chmod +s su
// > ./su
// Password:
```

Case study

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- `grep -r` is your friend!

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- Coreutils project
- OpenBSD source code (heavily recommended)
- `sudo-project/sudo`

Questions?

References:

- The Linux Programming Interface - Chapter 9
- `chmod(1)`